Handling Missing Data - fillna, interpolate, dropna

In [26]:

```
import pandas as pd
df = pd.read_csv("weather_data.csv",parse_dates=['day'])
type(df.day[0])
df
```

Out[26]:

	day	temperature	windspeed	event
0	2017-01-01	32.0	6.0	Rain
1	2017-01-04	NaN	7.0	Sunny
2	2017-01-05	28.0	NaN	Snow
3	2017-01-06	NaN	7.0	NaN
4	2017-01-07	32.0	NaN	Rain
5	2017-01-08	NaN	NaN	Sunny
6	2017-01-09	NaN	NaN	NaN
7	2017-01-10	34.0	8.0	Cloudy
8	2017-01-11	40.0	12.0	Sunny

In [27]:

```
df.set_index('day',inplace=True)
df
```

Out[27]:

	temperature	windspeed	event
day			
2017-01-01	32.0	6.0	Rain
2017-01-04	NaN	7.0	Sunny
2017-01-05	28.0	NaN	Snow
2017-01-06	NaN	7.0	NaN
2017-01-07	32.0	NaN	Rain
2017-01-08	NaN	NaN	Sunny
2017-01-09	NaN	NaN	NaN
2017-01-10	34.0	8.0	Cloudy
2017-01-11	40.0	12.0	Sunny

fillna

Fill all NaN with one specific value

In [28]:
```
new_df = df.fillna(0)
new_df
```

Out[28]:

day	temperature	windspeed	event
2017-01-01	32.0	6.0	Rain
2017-01-04	0.0	7.0	Sunny
2017-01-05	28.0	0.0	Snow
2017-01-06	0.0	7.0	0
2017-01-07	32.0	0.0	Rain
2017-01-08	0.0	0.0	Sunny
2017-01-09	0.0	0.0	0
2017-01-10	34.0	8.0	Cloudy
2017-01-11	40.0	12.0	Sunny

Fill na using column names and dict

In [29]:
```
new_df = df.fillna({
        'temperature': 0,
        'windspeed': 0,
        'event': 'No Event'
    })
new_df
```

Out[29]:

day	temperature	windspeed	event
2017-01-01	32.0	6.0	Rain
2017-01-04	0.0	7.0	Sunny
2017-01-05	28.0	0.0	Snow
2017-01-06	0.0	7.0	No Event
2017-01-07	32.0	0.0	Rain
2017-01-08	0.0	0.0	Sunny
2017-01-09	0.0	0.0	No Event
2017-01-10	34.0	8.0	Cloudy

2017-01-11	40.0	12.0	Sunny

Use method to determine how to fill na values

In [60]:
```
new_df = df.fillna(method="ffill")
new_df
```

Out[60]:

	temperature	windspeed	event
day			
2017-01-01	32.0	6.0	Rain
2017-01-04	32.0	7.0	Sunny
2017-01-05	28.0	7.0	Snow
2017-01-06	28.0	7.0	Snow
2017-01-07	32.0	7.0	Rain
2017-01-08	32.0	7.0	Sunny
2017-01-09	32.0	7.0	Sunny
2017-01-10	34.0	8.0	Cloudy
2017-01-11	40.0	12.0	Sunny

In [62]:
```
new_df = df.fillna(method="bfill")
new_df
```

Out[62]:

	temperature	windspeed	event
day			
2017-01-01	32.0	6.0	Rain
2017-01-04	28.0	7.0	Sunny
2017-01-05	28.0	7.0	Snow
2017-01-06	32.0	7.0	Rain
2017-01-07	32.0	8.0	Rain
2017-01-08	34.0	8.0	Sunny
2017-01-09	34.0	8.0	Cloudy
2017-01-10	34.0	8.0	Cloudy
2017-01-11	40.0	12.0	Sunny

Use of axis

In [66]:
```
new_df = df.fillna(method="bfill", axis="columns") # axis is either "index" or "
```

```
new_df = df.fillna(method="bfill", axis="columns") # axis is either "index" or
new_df
```

Out[66]:

day	temperature	windspeed	event
2017-01-01	32	6	Rain
2017-01-04	7	7	Sunny
2017-01-05	28	Snow	Snow
2017-01-06	7	7	NaN
2017-01-07	32	Rain	Rain
2017-01-08	Sunny	Sunny	Sunny
2017-01-09	NaN	NaN	NaN
2017-01-10	34	8	Cloudy
2017-01-11	40	12	Sunny

limit parameter

In [40]:

```
new_df = df.fillna(method="ffill",limit=1)
new_df
```

Out[40]:

day	temperature	windspeed	event
2017-01-01	32.0	6.0	Rain
2017-01-04	32.0	7.0	Sunny
2017-01-05	28.0	7.0	Snow
2017-01-06	28.0	7.0	Snow
2017-01-07	32.0	7.0	Rain
2017-01-08	32.0	NaN	Sunny
2017-01-09	NaN	NaN	Sunny
2017-01-10	34.0	8.0	Cloudy
2017-01-11	40.0	12.0	Sunny

interpolate

In [43]:

```
new_df = df.interpolate()
new_df
```

Out[43]:

day	temperature	windspeed	event
2017-01-01	32.000000	6.00	Rain
2017-01-04	30.000000	7.00	Sunny
2017-01-05	28.000000	7.00	Snow
2017-01-06	30.000000	7.00	NaN
2017-01-07	32.000000	7.25	Rain
2017-01-08	32.666667	7.50	Sunny
2017-01-09	33.333333	7.75	NaN
2017-01-10	34.000000	8.00	Cloudy
2017-01-11	40.000000	12.00	Sunny

In [33]:

```
new_df = df.interpolate(method="time")
new_df
```

Out[33]:

day	temperature	windspeed	event
2017-01-01	32.000000	6.00	Rain
2017-01-04	29.000000	7.00	Sunny
2017-01-05	28.000000	7.00	Snow
2017-01-06	30.000000	7.00	NaN
2017-01-07	32.000000	7.25	Rain
2017-01-08	32.666667	7.50	Sunny
2017-01-09	33.333333	7.75	NaN
2017-01-10	34.000000	8.00	Cloudy
2017-01-11	40.000000	12.00	Sunny

Notice that in above temperature on 2017-01-04 is 29 instead of 30 (in plain linear interpolate)

There are many other methods for interpolation such as quadratic, piecewise_polynomial, cubic etc. Just google "dataframe interpolate" to see complete documentation

dropna

In [34]:

```
new_df = df.dropna()
new_df
```

Out[34]:

day	temperature	windspeed	event
2017-01-01	32.0	6.0	Rain
2017-01-10	34.0	8.0	Cloudy
2017-01-11	40.0	12.0	Sunny

In [45]:

```
new_df = df.dropna(how='all')
new_df
```

Out[45]:

day	temperature	windspeed	event
2017-01-01	32.0	6.0	Rain
2017-01-04	NaN	7.0	Sunny
2017-01-05	28.0	NaN	Snow
2017-01-06	NaN	7.0	NaN
2017-01-07	32.0	NaN	Rain
2017-01-08	NaN	NaN	Sunny
2017-01-10	34.0	8.0	Cloudy
2017-01-11	40.0	12.0	Sunny

In [49]:

```
new_df = df.dropna(thresh=1)
new_df
```

Out[49]:

day	temperature	windspeed	event
2017-01-01	32.0	6.0	Rain
2017-01-04	NaN	7.0	Sunny
2017-01-05	28.0	NaN	Snow
2017-01-06	NaN	7.0	NaN
2017-01-07	32.0	NaN	Rain
2017-01-08	NaN	NaN	Sunny
2017-01-10	34.0	8.0	Cloudy
2017-01-11	40.0	12.0	Sunny

Inserting Missing Dates

In [56]:

```
dt = pd.date_range("01-01-2017","01-11-2017")
idx = pd.DatetimeIndex(dt)
df.reindex(idx)
```

Out[56]:

	temperature	windspeed	event
2017-01-01	32.0	6.0	Rain
2017-01-02	NaN	NaN	NaN
2017-01-03	NaN	NaN	NaN
2017-01-04	NaN	7.0	Sunny
2017-01-05	28.0	NaN	Snow
2017-01-06	NaN	7.0	NaN
2017-01-07	32.0	NaN	Rain
2017-01-08	NaN	NaN	Sunny
2017-01-09	NaN	NaN	NaN
2017-01-10	34.0	8.0	Cloudy
2017-01-11	40.0	12.0	Sunny